Demographics for Civilization VII

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Figure 1: Demographics logo

Demographics

A Civilization VII mod. Open the Demographics screen at any point in a game and see how every civilization has done since the first turn: per-turn graphs of yields, economy, land and military, rankings of civilizations and settlements, the diplomacy between them, every war and crisis, and the story of your campaign told age by age. A Hall of Fame ranks every campaign you have played, and opens from the main menu too.

The mod samples every civilization each turn and builds everything it shows from those samples. It only reads the game: no yields, balance or AI behavior are changed.

Gold per turn for every civilization across Antiquity and Exploration, with the crisis stages, the wonders and the age



Figure 2: Gold per turn for every civilization across Antiquity and Exploration, with crisis stages, wonders, and the age change marked on the timeline

change marked on the timeline.

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What the player does

Open the screen from the Demographics button in the subsystem dock, from the pause menu, or from the end-of-game screen, where it opens on the Hall of Fame. The **Hall of Fame** button on the main menu shows past campaigns with no game loaded. **Escape** closes either one.

The screen has three rows of navigation. The top tabs pick a hub: **Global Statistics**, **Geopolitics**, **History** and **World Rankings**, plus **Emigration** when that companion mod is installed. The page tabs under them pick a page, such as *Yields Per Turn* or *Military Power*, and the metric pills pick the chart on that page.

Every chart page shares one toolbar: a time range (25 to 1000 years, the current age, a single age, or all time), the label order (**Civ → Leader** or **Leader → Civ**), the time-axis mode, wonder markers on or off, and **Copy as CSV**. Charts mark each crisis stage and age change with a vertical line, and each wonder with its icon on the line of the civilization that built it. Clicking a civilization in the legend hides or shows it.

The **Options** button and the active analytics policy sit together at the top right of every tab, level with the title. The button opens the game's own Options screen at **Mods → Demographics**, where every setting lives.

Global Statistics

Per-turn history for every civilization you have met, on five pages:

- **Yields Per Turn:** Gold, Production, Food, Science, Culture, Influence and Happiness per turn.
- **Economy & Resources:** GDP, Treasury, Trade Routes, Total Resources, a stacked chart of one civilization's resources by type, and separate Bonus, Empire and City resource counts, plus Treasure resources in Exploration and Factory resources in the Modern age.
- **Society:** Great Works, Techs, Civics, Wonders, a **Wonders Board** listing each civilization's wonders with the turn they were built, **Wonder Races** showing who built each wonder first, Natural Wonders Discovered, and a **Triumphs Radar** of Legacy Path progress. Faith, Tourism and Great People appear once any civilization has recorded a value for them.
- **Religion:** in Antiquity, the pantheons each civilization chose and their yields; from Exploration on, religion standings (settlements following each religion), religious spread over time, and followers by population.
- **Land & Settlements:** Land Area, each civilization's share of the world's land, settlements over time, a settlement-size distribution with the most urbanized settlements, settlement-cap use and the cap over time, city and town counts, and boards of quarters and buildings constructed by type.

Without *Emigration*, **Population** sits on the Society page. With it, Population and Population Share move to the Emigration tab's first page, beside the migration charts.

Geopolitics

- **Global Relations:** a ring of your civilization and the civilizations you have met, switchable to **City States**. Its lines show wars, alliances, open borders, denouncements and attitudes; a second view, **Agreements**, draws the active agreements between civilizations.
- **Agreements:** Diplomatic Approval and Ongoing Deals over time.
- **Crises: Crisis Impact**, the cost of each crisis stage per civilization, and **Crisis Impact Graphs**, each crisis statistic charted per civilization.
- **Soft Power:** Score, the **Power Race** (every civilization's score rank, turn by turn), three scatter plots (Power & Science, Wealth & Culture, Happiness & Influence), and an **Archetype** radar comparing science, military, economy, culture, people and land.
- **Military Power:** Military Power, Units Killed, Units Lost, Battles, Wars Declared, Wars Received, Settlements Conquered, Conquest %, the **War Timeline**, **War Impact**, and boards of units trained, kills and losses by unit type.

Wars and crises

The **War Timeline** is a Gantt chart of every war, built from diplomacy events: the declarer, the supporters and the opposers share one war, so a coalition reads as a single conflict, and each war gets a generated name. **War Impact** picks one war and charts each side through it: military strength, units lost, production directed to war, settlements and land lost or gained, settlements razed, population lost, and more.

Crisis Impact splits each age's crisis into its stages (Begins, Intensifies, Culminates, Ends), each with a cost table per civilization, then a cumulative table for the whole crisis and, once crises span more than one age, a total across ages.

History

Three pages, recorded from the moment Demographics is active in a game. When the mod is added to a game already in progress, the chronicle starts from that turn and the first age's summary says so.

Chronicle tells the campaign's history age by age. Each age opens with a summary of your civilization's deeds (settlements founded, captured and lost, wonders, Triumphs, wars, the religion you founded, civilizations destroyed), followed by every recorded event in order: settlements founded, taken or lost, wonders completed, wars and peace, religions founded, Triumphs earned, crisis stages, natural disasters that struck a civilization's land, first contacts, eliminations, new ages and victories. Pills filter by age and by kind of event, and the order can run newest or oldest first. Events involving civilizations you have not met are hidden and counted, following the analytics policy.

Timeline puts the whole game on one scrolling chart. A banner for each age carries your civilization's emblem and the age's dates. Below it run lanes for your wars (in the enemy's color, with their emblem) and the age crises stage by stage, then one lane per kind of milestone (**Wonders**, **Triumphs**, **Religion**, **Conquests**, **Victory & falls**) drawn with the game's own icons, then every settlement founded, natural disasters, migration in and out (with the *Emigration* mod), the population curve with its milestones, and a turn ruler.

Pills show all ages or one, zoom runs from 1× to 8× with pan buttons, clicking the ruler moves the cursor, and **Play** sweeps through the game, naming each event as it passes. Everything on the chart has a tooltip, and hovering the population lane or the ruler reads out the turn and population under the mouse.

The row above the timeline picks whose story it shows. Every game opens on your own civilization; add any civilization you met, or **All**, to put their milestones, settlements, wars with each other, disasters on their land and population line on the same lanes.

Above the timeline, a territory map lays the world out like the game's own minimap, one cell per tile: terrain by biome, each civilization's land tinted in its color, independent land in gray, settlements marked, and land you had not yet explored left blank. It follows the timeline's cursor, so playback replays both the expansion and your exploration. Civilizations you have not met stay off the map.

Lineage has one row per leader and one column per age: each cell is the civilization that leader led in that age, in the leader's color, with the turn a line ended.

World Rankings

- **Civilization Ranking:** a podium of the three strongest civilizations, each with its capital and **View on map** / **Cinematic view** buttons, and the full list ordered by cumulative settlement score, with population and settlement counts.
- **Civilization Rank by Yield:** every civilization's rank, or its value with **Rank / Value**, in each tracked figure, in two tables that share one sort: the rates and yields first, then **Totals & Tallies** (land, cities and towns, conquests, units, great works, migration). Click any column to sort both.
- **Top 25 Settlements:** a podium and ranked list of the strongest settlements. Each row shows the settlement's wonders (hover one for the year it was completed), a **Holy City** badge with the religion's icon when it is a religion's holy city, and small icons under the rank for every output it leads the world in. **View on map** snaps the camera to the city and **Cinematic view** orbits it; the experimental flyby, on by default in Options, adds a short keyframed flight. Once an age ends, a pill for it (for example **End of Antiquity**) shows that age's final top ten.
- **Settlement Rank by Yield:** a sortable table of every settlement's population and yields, filterable to **All**, **Cities** or **Towns**.
- **Hall of Fame:** every campaign played on this computer (below).

Across the first four, places 1–3 carry a gold, silver or bronze wash, your own civilization and settlements are outlined in gold, and in the two tables the cell of whoever leads each column is highlighted in gold (hover it for the category). Civilizations you have not met show as placeholders unless name revelation is enabled in Options.

Hall of Fame

The Hall of Fame lists every campaign played on this computer, the current one included as it is played. The same view opens from the **Hall of Fame** button on the main menu, with no game loaded.

Best Games opens first: your totals, a podium of your three best games, the rest of your top ten, and the game you are playing at its rank between the games just above and below it (from the main menu, your most recent game), with victories by type. **Rankings** orders the games by result (victories first), then Triumphs earned across every age, then fewest turns, and gives each an honorific from Augustus Caesar down to Ethelred the Unready.

Leaders and **Civilizations** give each one a win rate over finished games, drawn as a bar that is green for the games won and red for the rest, with the attempts, most Triumphs and average length beside it. **Records** names the game that holds each record: most Triumphs, most wonders, largest empire, most settlements taken, fastest victory, longest game.

Any game opens a page with its lineage and figures, the territory map beside its rivals, the game's timeline with playback, and its highlights age by age, where each Triumph says what it was earned for and what it gave. **Game options** at the foot of that page removes the game from the Hall of Fame.

Hide short games / Show all games sits under the section tabs on every page. Unfinished games under 20 turns are hidden by default as test loads, and the row says how many it is hiding.

Screenshots



Figure 3: GDP over time, shown in real-world dollars

Every view of the screen, the ones above and the rest, is in docs/screenshots and attached to each GitHub release.



Figure 4: Resources held, stacked by type

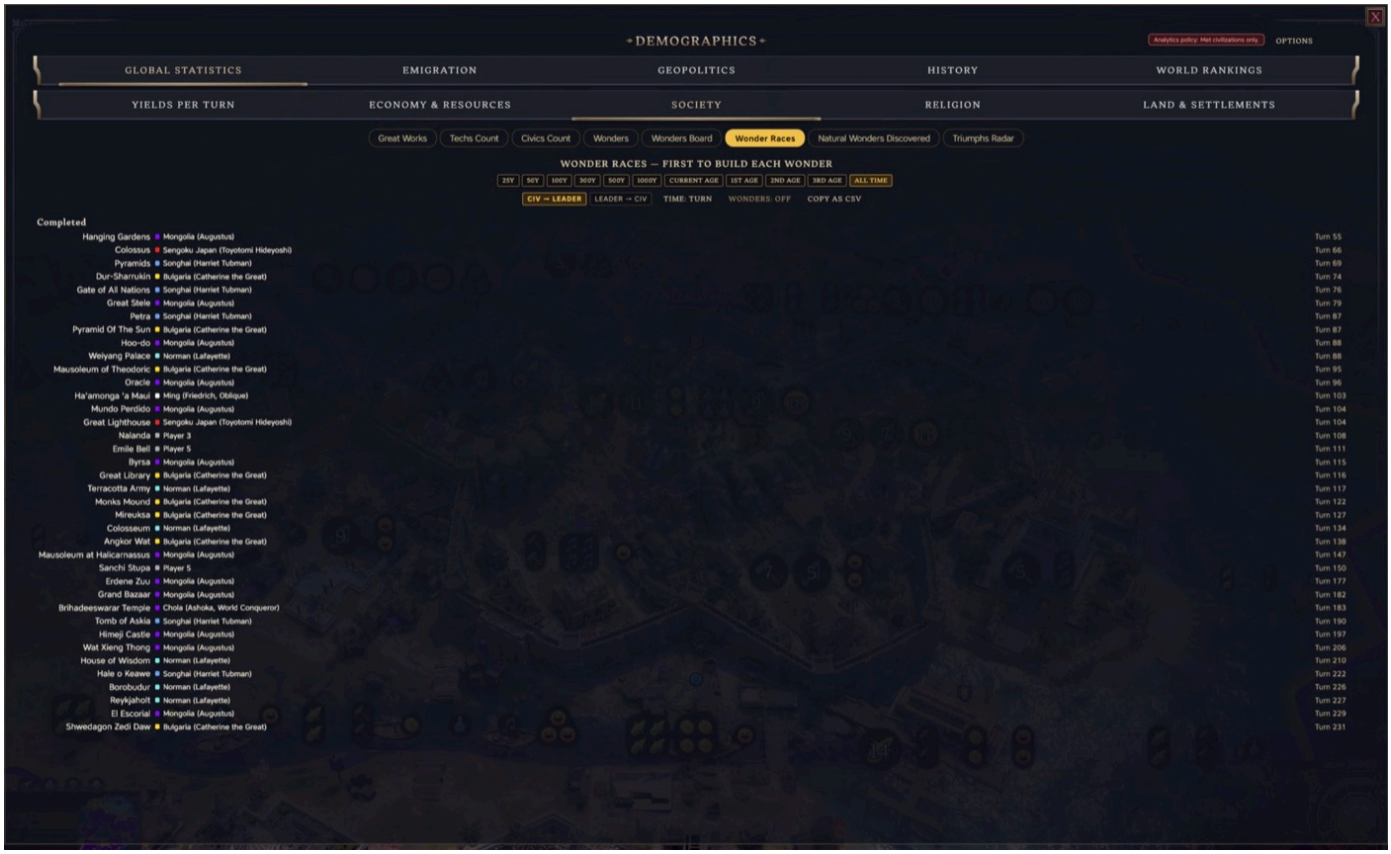


Figure 5: Wonder Races: who built each wonder first, and when

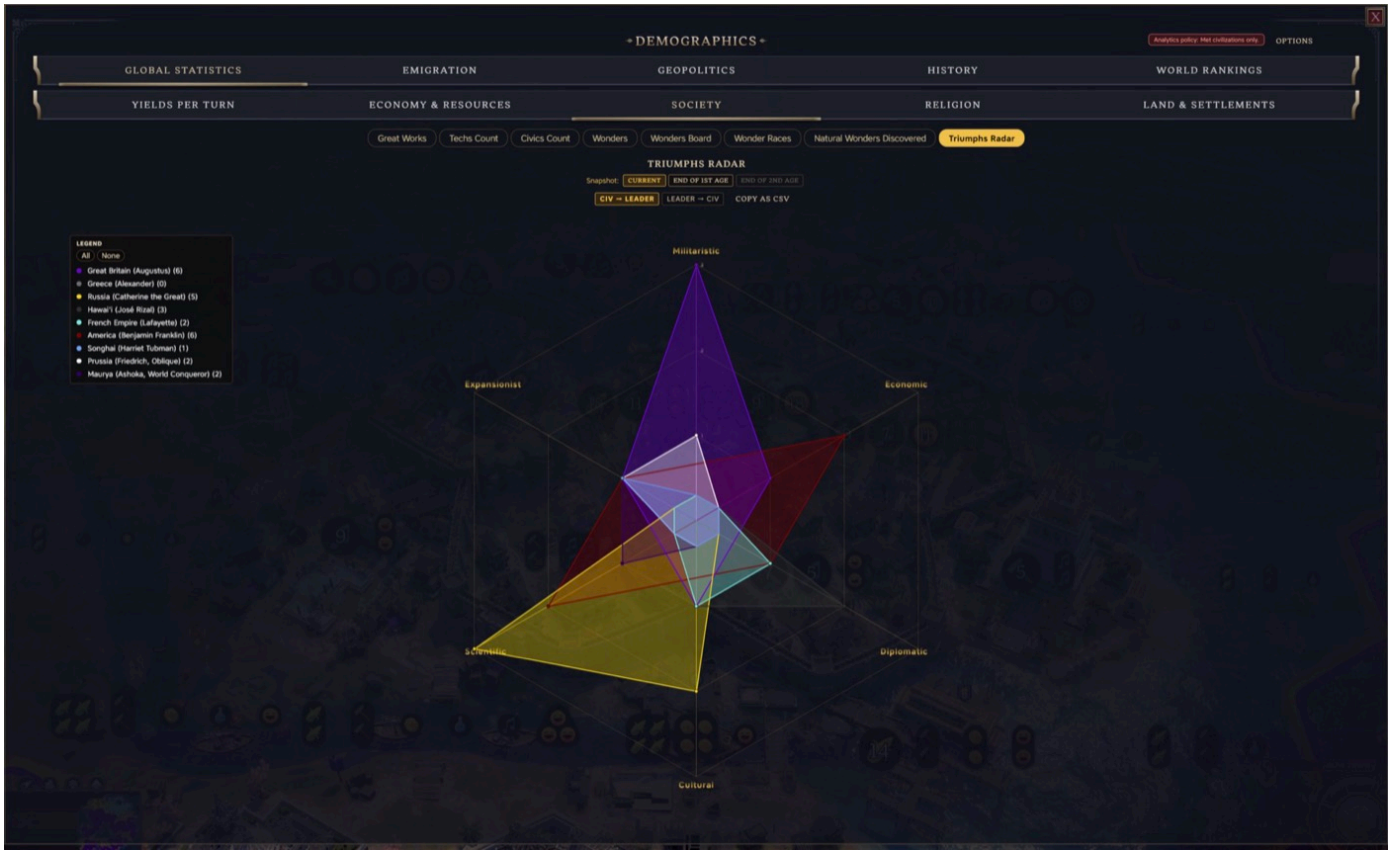


Figure 6: Triumphs radar: Legacy Path progress side by side



Figure 7: Share of the world's land over time



Figure 8: Settlement sizes and the most urbanized settlements

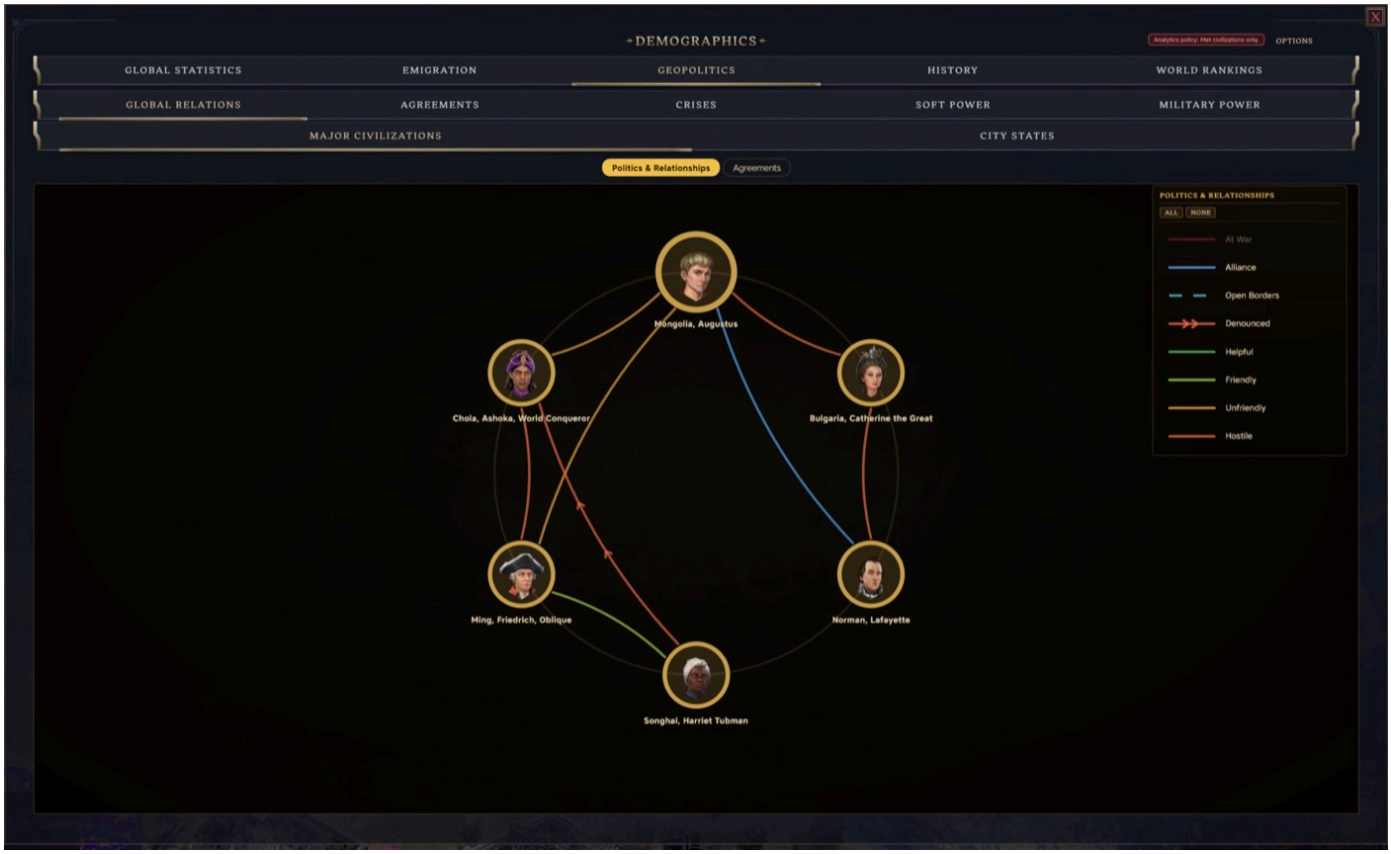


Figure 9: Global Relations: wars, alliances and attitudes between met civilizations

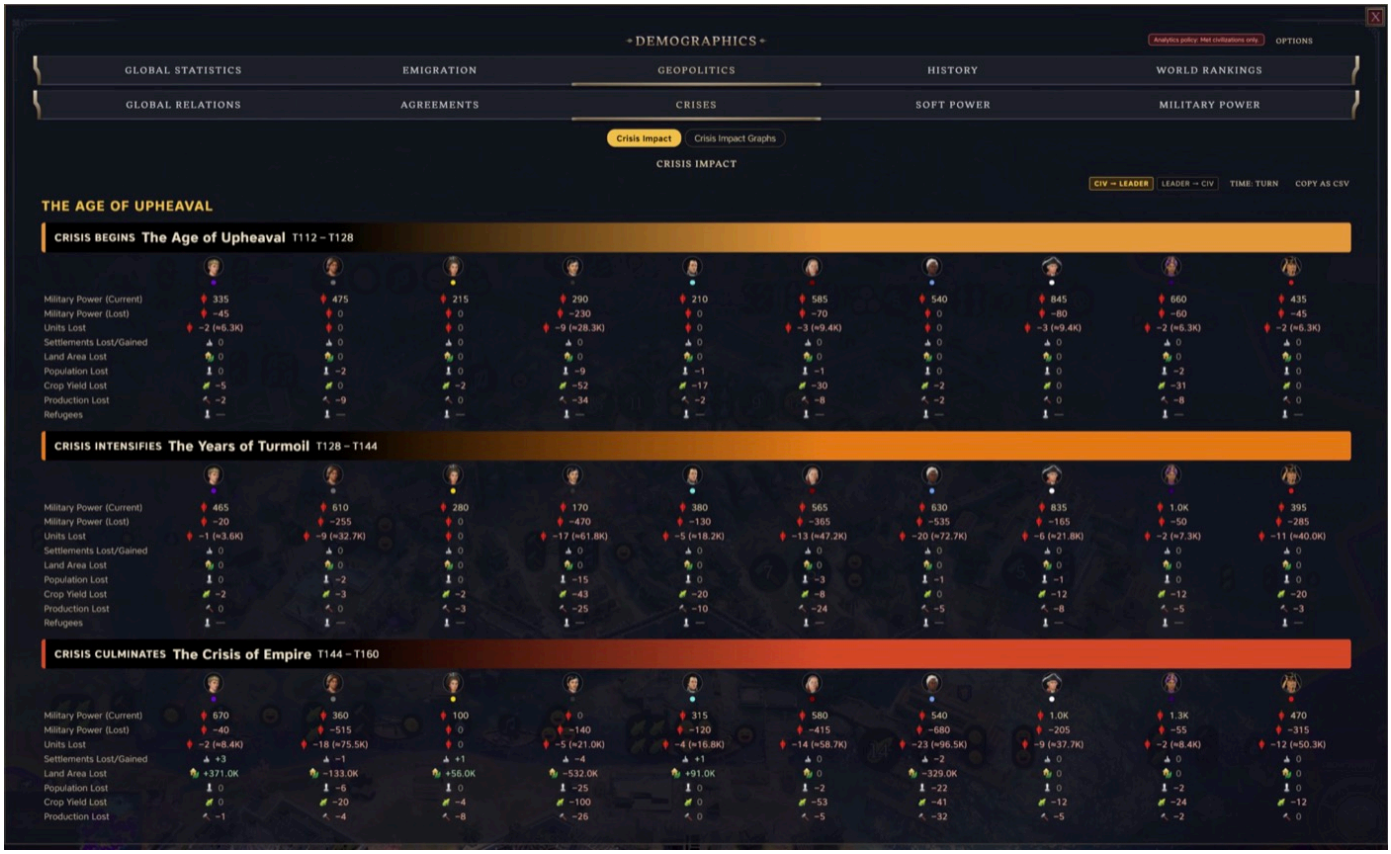


Figure 10: Crisis Impact: what each stage of the crisis cost every civilization



Figure 11: Power Race: every civilization's score rank, turn by turn

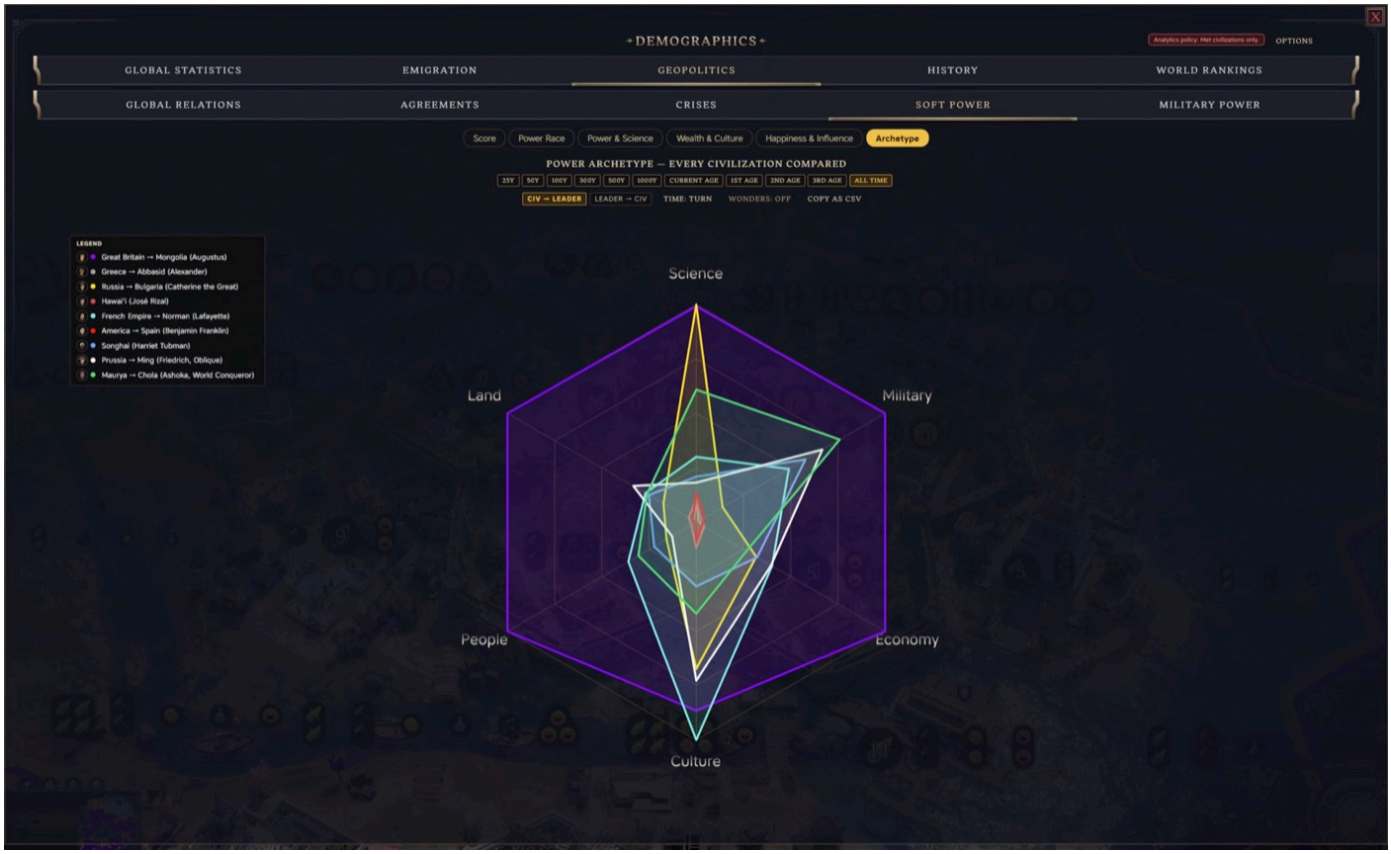


Figure 12: Archetype: science, military, economy, culture, people and land compared



Figure 13: War Timeline: every war this game, with its generated name



Figure 14: Civilization Ranking

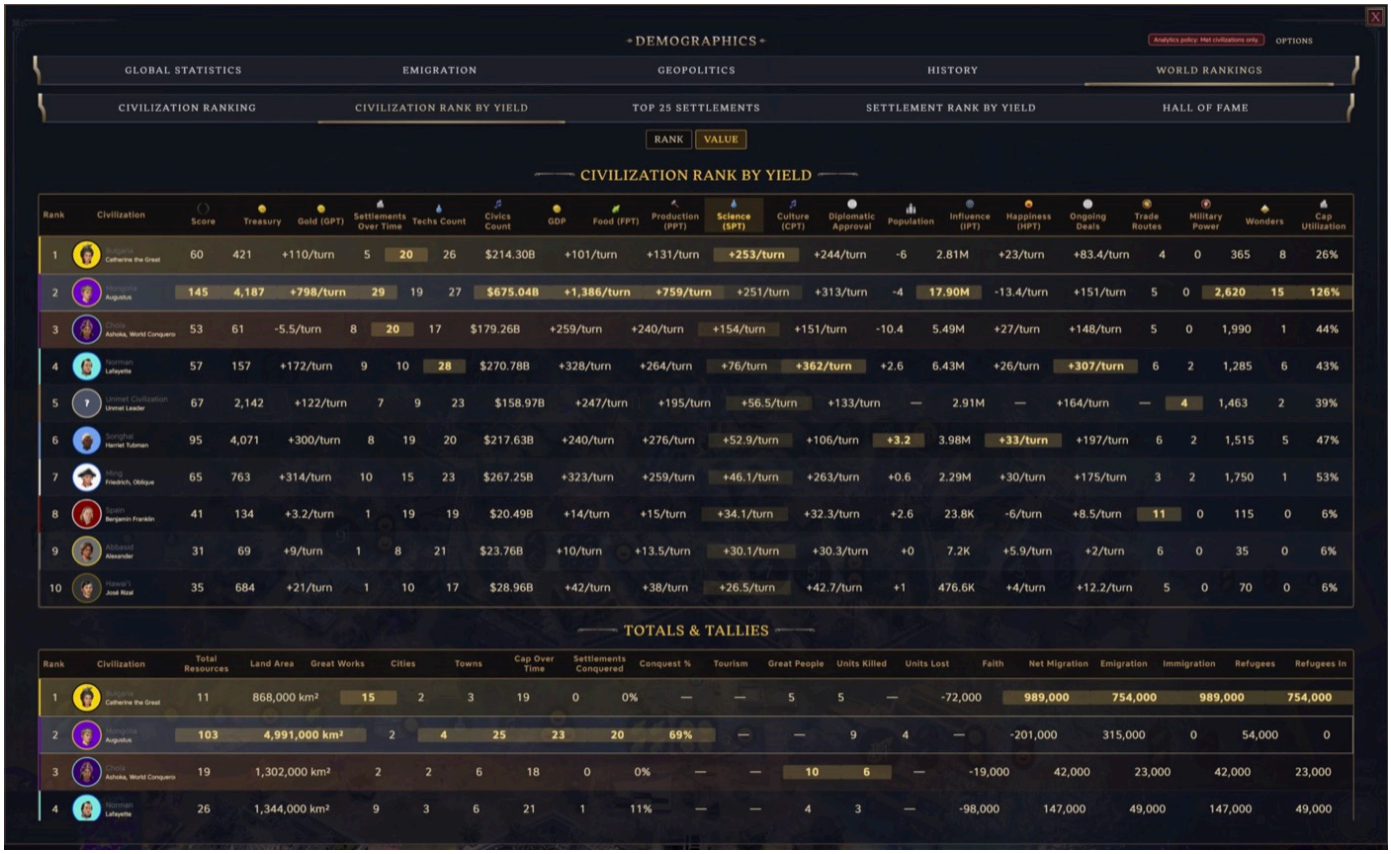


Figure 15: Civilization Rank by Yield, with Totals & Tallies below

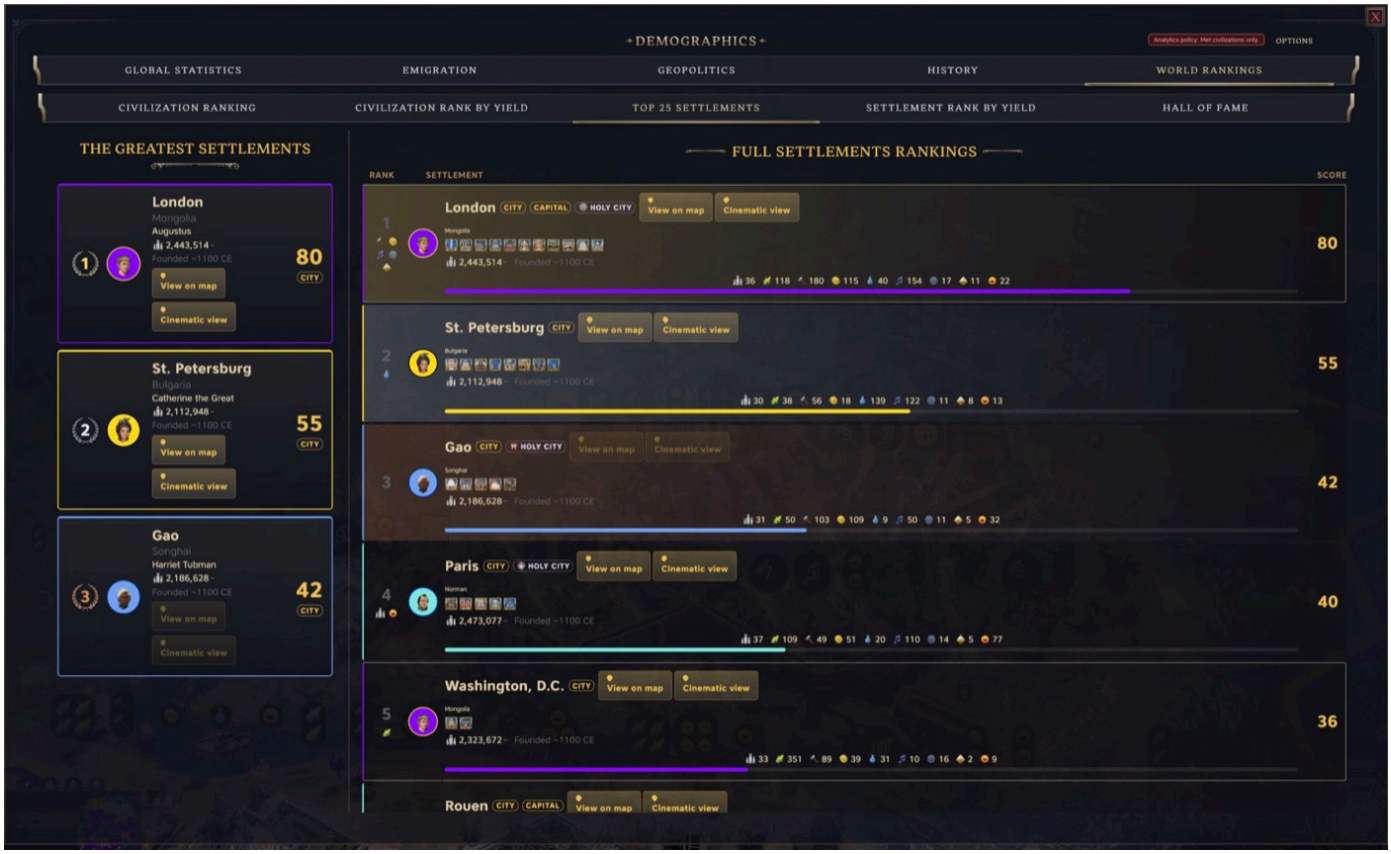


Figure 16: Top 25 Settlements: holy cities, world leaders and map fly-to

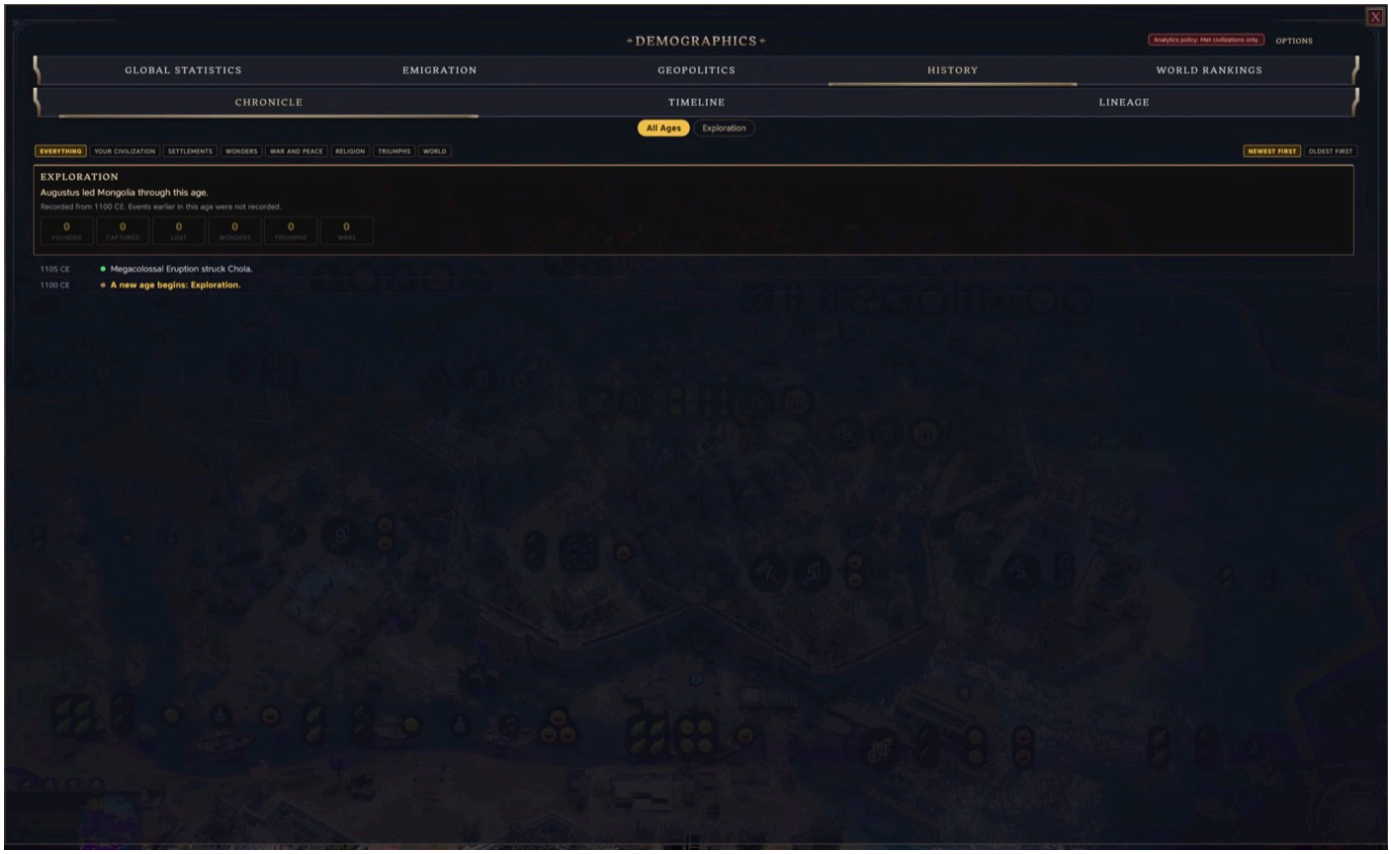


Figure 17: Chronicle: your campaign told age by age

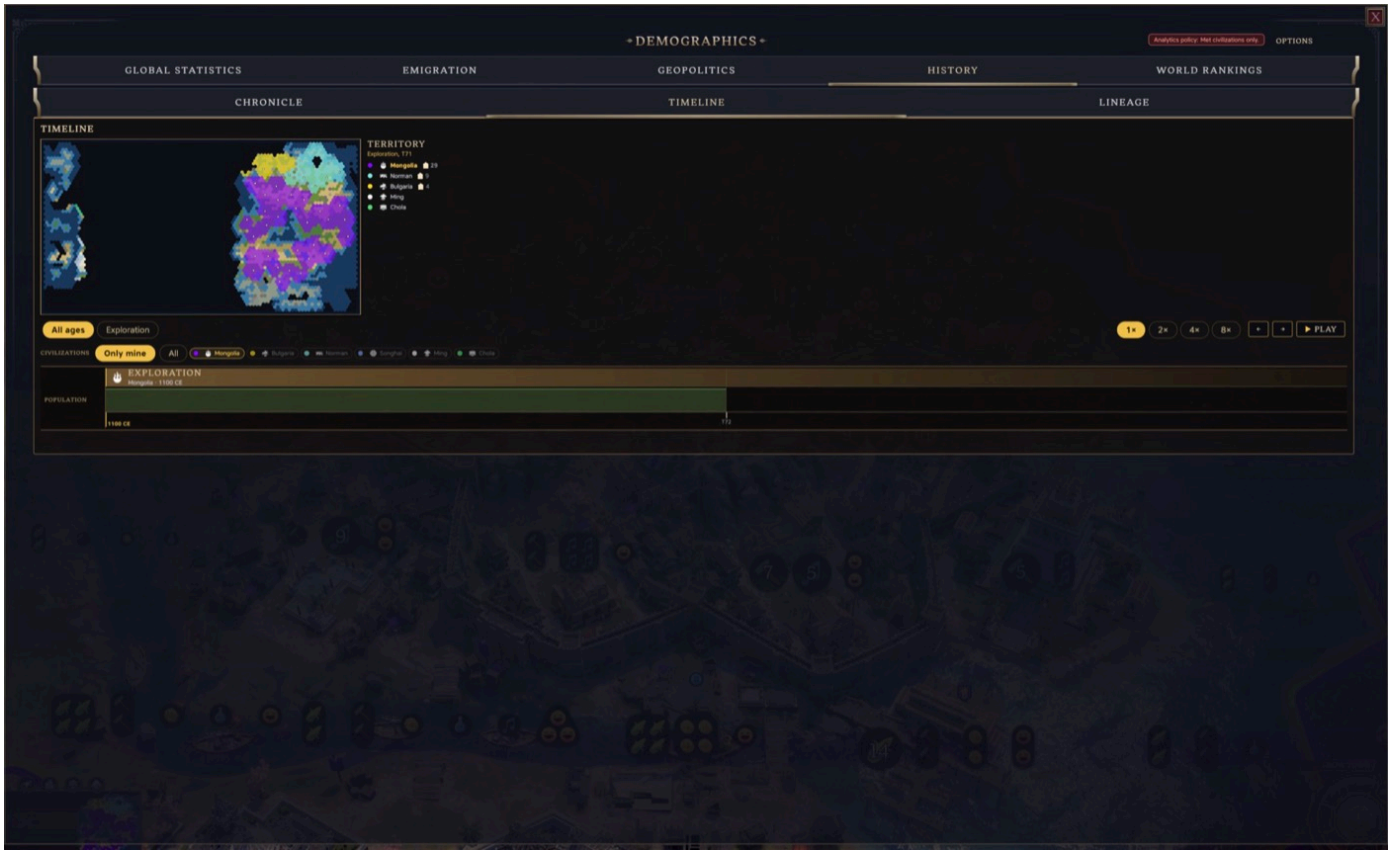


Figure 18: Timeline: wars, crises, wonders, Triumphs, settlements, disasters and population, with the territory map

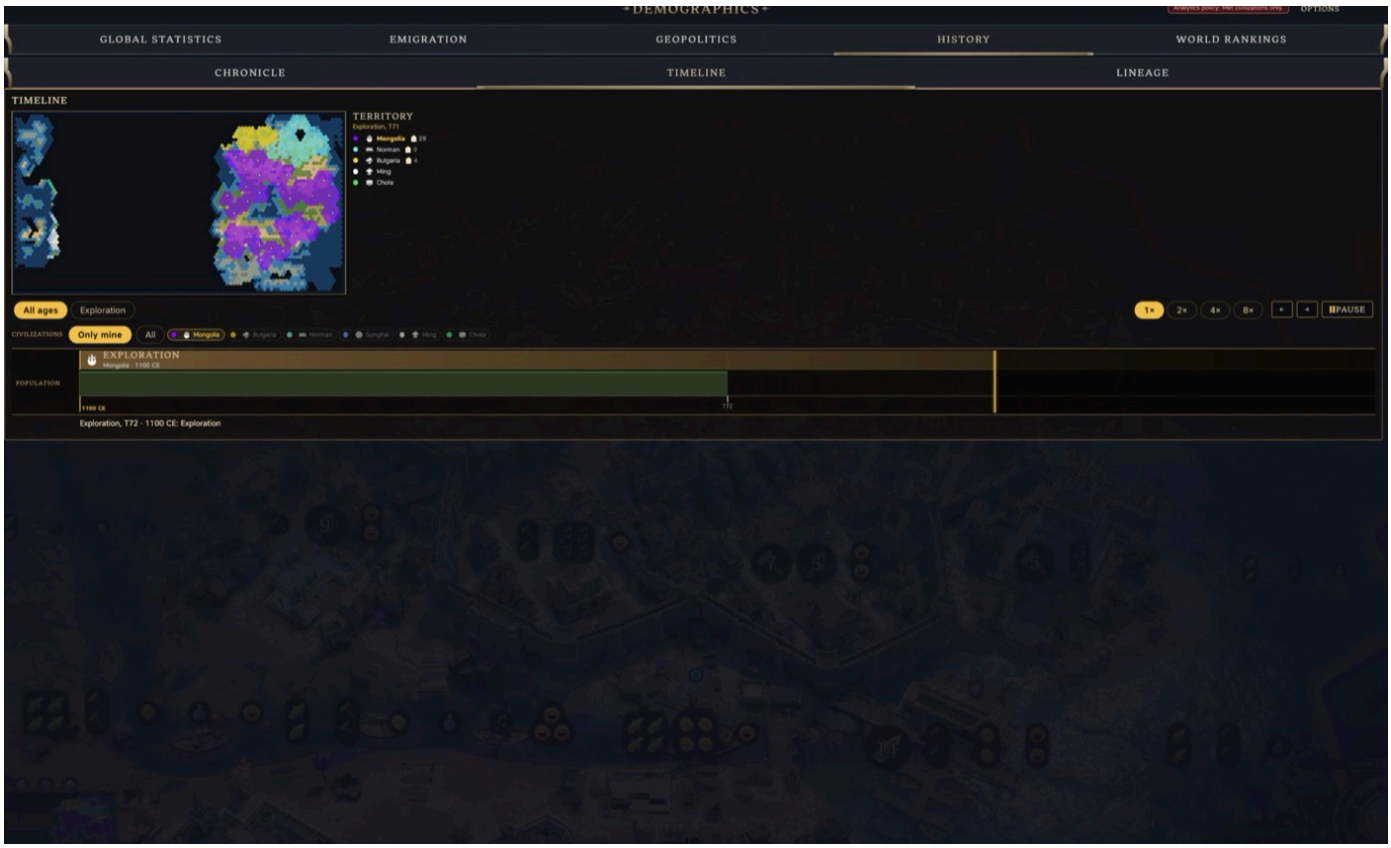


Figure 19: Play replays the game: the map and caption follow the cursor



Figure 20: Lineage: every leader and the civilizations they led



Figure 21: Best Games: your finest playthroughs, and this game in context

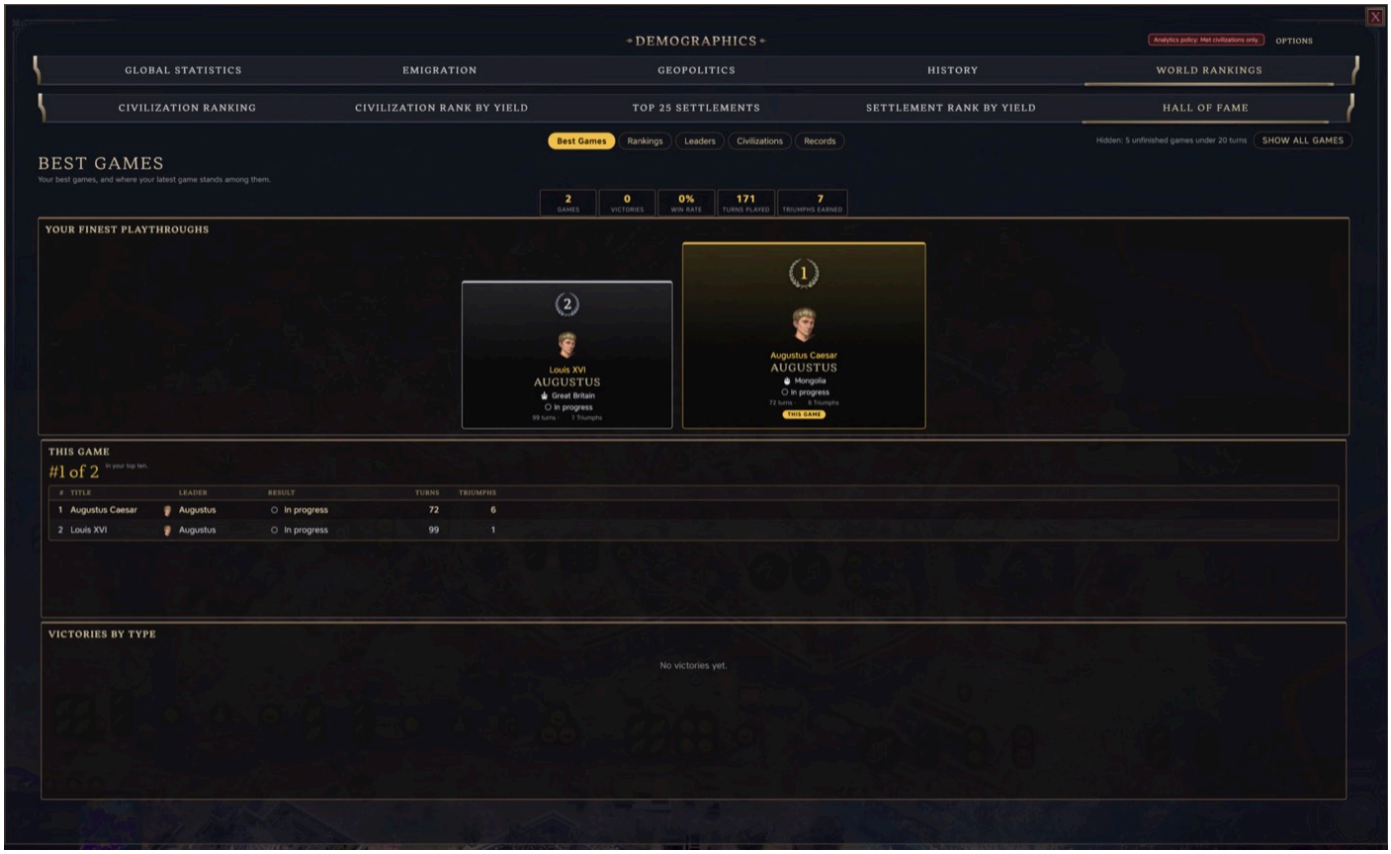


Figure 22: A game's page in the Hall of Fame

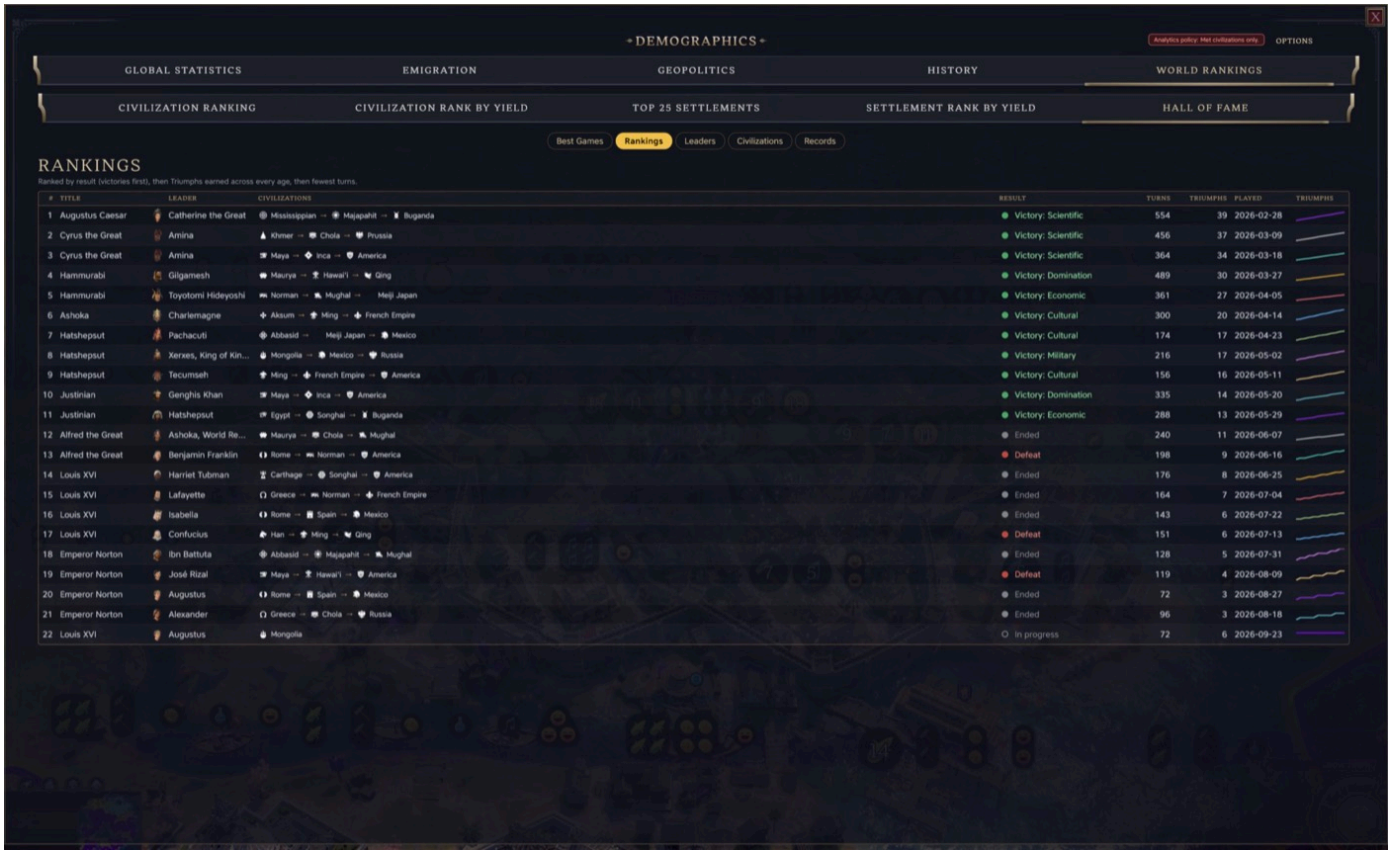


Figure 23: Rankings, each game with its honorific

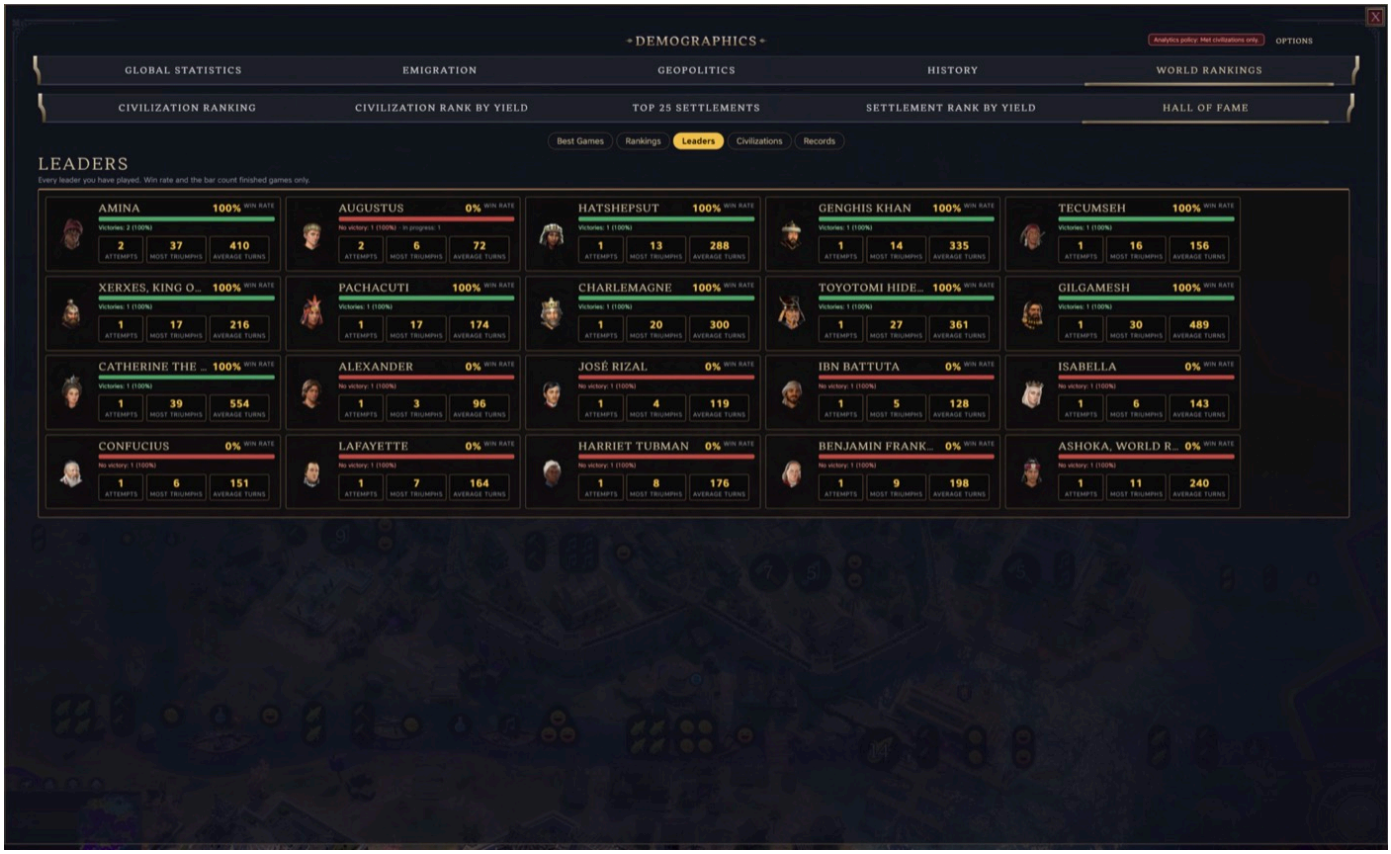


Figure 24: Leaders you have played

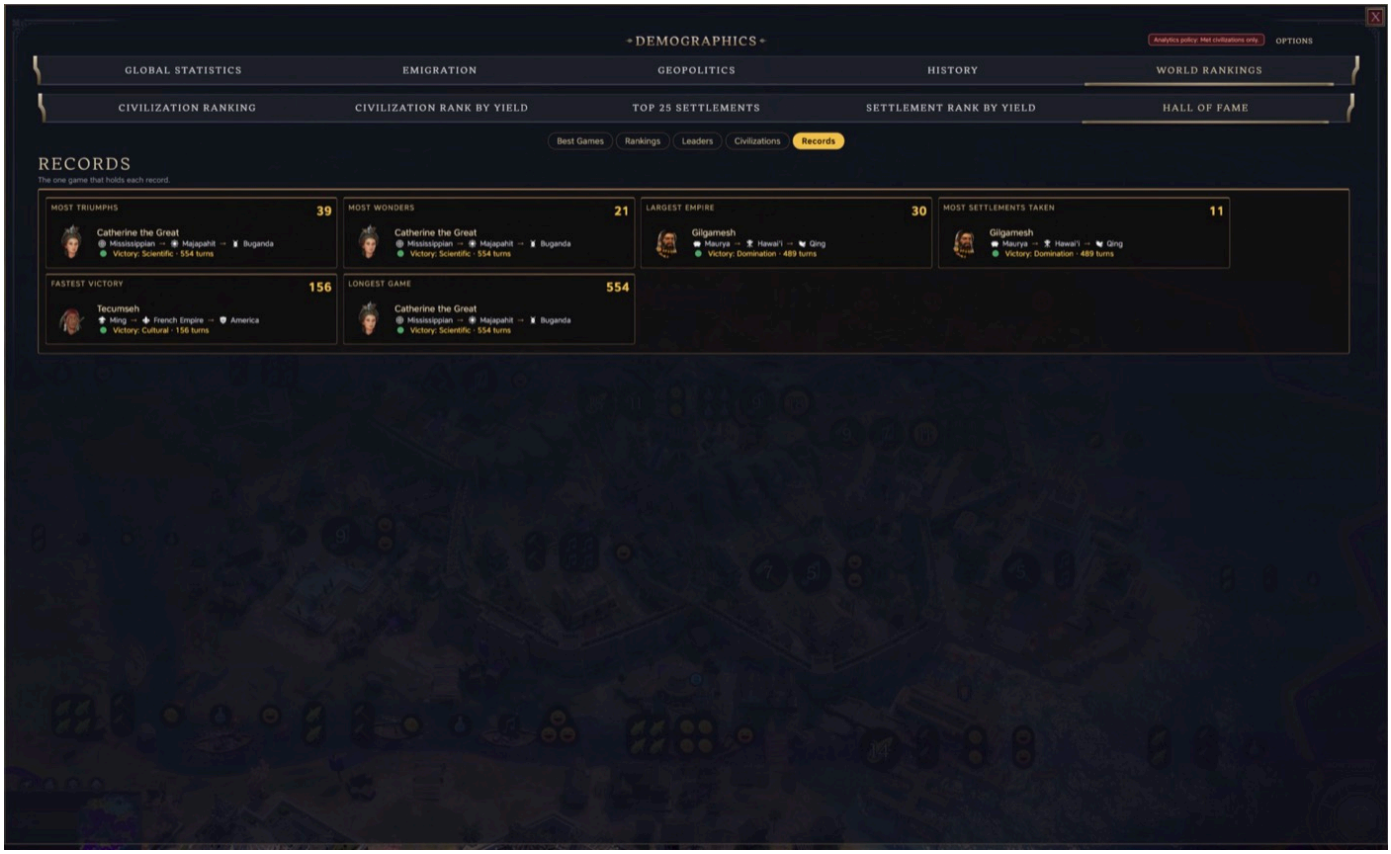


Figure 25: Records: the game that holds each one



Figure 26: Cinematic view: the camera tours a settlement's wonders and quarters

Settings

Every setting is in the game's Options screen under **Mods → Demographics**.

- **Spoiler guard** (on by default) hides the names, charts, and diplomacy and relations figures of civilizations you have not met: they show as placeholders, and their lines show a gap. **History on first contact** chooses whether meeting a civilization reveals its whole history or only what follows the meeting. A banner below the views states the analytics policy in force.
- **Complexity** sets how much of the screen is shown. **Basic** shows the core pages: Yields Per Turn, Economy & Resources, Society, Land & Settlements, Population, Global Relations and Military Power. **Standard** (the default) shows every page, adding Religion, Agreements, Crises and Soft Power.
- **Colorblind mode** swaps chart and relationship colors for a colorblind-safe set.
- **Smooth chart lines** draws history lines as a 3-turn moving average; the recorded data is unchanged.
- Show eliminated civilizations; wonder, war and disaster markers on the charts; and the cinematic view, the flyby, its length and whether the camera may rotate.

The game's own **Font Size** setting scales every page of the screen.

Display scaling

The screen is drawn as one uniformly scaled frame against a 2880x1800 reference, so the same layout reaches every display rather than filling the screen at one resolution and crowding it at another. The scale has a lower bound and text has its own, higher one, so nothing shrinks past reading size. 2880x1800 renders at full size; 3840x2160, 2560x1440, 1920x1080, 1600x900 and 1366x768 approximate it.

Compatibility

- Does not replace base-game files and changes no gameplay data.

- Every read from the game is defensive: a missing or changed field means missing data, not a crash. A view that fails says "Chart failed to render" in its own panel, and the rest of the screen keeps working.
- The mod's CSS custom properties start with `--dg-`, so it styles only its own screen and leaves other mods' variables alone.
- Recorded history is saved in each save game, through the game configuration store, and carries across quitting, loading and age transitions. If that store is unavailable, history is kept for the session only.
- Multiplayer: the host's analytics policy is a ceiling for every player in the game, and the banner says when the host's policy, rather than your own setting, is what hides a civilization. In an internet, LAN or cloud game only the host writes the History campaign into the shared game configuration; guests keep it in memory and, on loading, take the host's copy with their own civilization as the viewpoint. Hotseat and single-player games store it as usual.
- With *Emigration*: Demographics hosts its tab and charts. If late-game turns feel heavy with both mods on, raise Emigration's *How often migration runs* (Options → Mods → Emigration → Advanced, Pacing group). It changes how often migration updates, not what it does. Demographics' own sampling adapts to the machine and the game by itself.
- Carries the Tower Settings Keeper, so its settings, and other mods' settings in the shared entry, survive a restart (below).
- Translated into German, Spanish, French, Italian, Japanese, Korean, Polish, Brazilian Portuguese, Russian and Simplified Chinese, with each language's own number formatting.

The Civilization VII storage bug

Civilization VII 1.5.0 has a bug in the storage mods use for their settings: whatever entry a mod asks for, the game returns whichever mod's entry sorts first. Only that one mod's data works. The entry that Demographics and most mod option panels share is never read back, so settings reset every launch and the Hall of Fame cannot read or save its list of past games (every game is still stored in its own save). It is a game bug for Firaxis to fix, and a reproduction is written up for their support portal.

Since 2.8.1 the mod carries the Tower Settings Keeper, which keeps every mod's settings inside the one entry the game reads correctly and puts a store another mod has disturbed back in order, without deleting anything.

If the Hall of Fame still cannot read its list, a notice box on every Hall of Fame page names the bug, and clicking it opens **Repair mod storage**. The repair empties the storage and starts it again with one entry, the shared settings entry most mod option panels use, holding this mod's settings and past games. Being the only entry, it is what the game returns, so every mod that keeps its settings there loads and saves them normally from then on. A mod with a separate entry of its own is not helped: it could not read that entry before and still cannot.

The repair permanently deletes the saved data of the one mod the game was reading for everyone, and clears what other mods had written but could never read back. Saved games, the history inside them, the game's own settings and every mod's files are untouched. The sheet says all of this before the button, the button says what it deletes, and it takes two clicks with a Cancel. It is offered again if a mod later writes its own first-sorting entry. The whole story: docs/civ7-storage-bug.md.

Installation

1. Subscribe on the Steam Workshop, or download the zip from the latest release.
2. Unzip it so the `demographics` folder sits in the Civilization VII Mods directory:
   - Windows: `%localappdata%/Firaxis Games/Sid Meier's Civilization VII/Mods/`
   - macOS: `~/Library/Application Support/Civilization VII/Mods/`
3. Enable **Demographics** from Additional Content in-game.

How the figures are calculated

Every number comes from what the mod can see by sampling the game each turn, not from a hidden engine ledger. Each turn it records every civilization's raw figures, and the charts and tables compute everything from those snapshots. Two things follow: a figure is only as fine-grained as the sampling, and losses are inferred from how a value moved over time rather than read from a casualty log.

Direct reads

Most figures are read straight off the player each turn and plotted as they are: the per-turn yields (Gold, Production, Science, Culture, Influence and Happiness per turn, net) and the counts (Settlements, Techs, Civics, Wonders, Trade Routes, Ongoing Deals).

Derived and rescaled figures

A few figures apply a fixed transform so the game's number reads at a believable real-world scale. These are cosmetic and never affect play.

- **Population (scaled):** derived from Civilization VII's own per-age growth formula (the food cost of growing a settlement, which differs by age) and turned into a count of people by one calibration constant. Each settlement is valued for the age it is in: towns in the thousands, great cities up to about a million before the Modern age, and megacities of 10 to 38 million only in the Modern age. The hand-off across an age change is continuous, and there is an upper safety bound. A small variation per settlement, drawn from its happiness, urban and rural mix and growth trend, keeps two settlements of the same size from reading identically. A civilization's figure is the sum of its settlements, so "empire" and "sum of cities" agree. The *Emigration* mod uses the same curve.
- **GDP:** a weighted sum of the per-turn yields, `× turn × 1,000,000`, shown in `$M` / `$B`. Weights: Gold 1.0, Production 1.0, Food 0.5, Science 1.2, Culture 1.2, Influence 1.5. Multiplying by the turn approximates a cumulative economy rather than one turn's output; the turn factor is capped at 300 so very long games and slow speeds keep a comparable scale.
- **Score:** `techs + civics + 2 × settlements + floor(gold ÷ 100)`, with techs and civics counted across ages so the line never drops at an age change. The **Power Race** plots each civilization's rank by this score.
- **Land Area:** `owned tiles × 7,000 km²`, a hex's nominal real-world area.
- **Military Power:** the summed combat strength of the civilization's military units.
- **Diplomatic Approval:** each met major civilization counts by relationship (Allied +5, Helpful +3, Friendly +2, Neutral 0, Unfriendly -2, Hostile -3, At War -5); city-states you are suzerain of count the same weights `× 0.3`.
- **Settlement Cap Utilization:** `settlements ÷ settlement cap × 100`.
- **Land Area Share** and **Population Share:** each civilization's land, or population, as a share of the total held by every charted civilization that turn, stacked to 100%.
- **Conquest %:** settlements a civilization has conquered, as a percentage of the settlements it holds now.
- **Crisis Stage:** the engine's stage (pre-crisis -1 through 3) shifted up by one, so it reads Stage 1 to Stage 4 and plots as a step.

War and crisis cost

War costs (the tables per combatant) and crisis costs (the stage and cumulative tables per civilization) are worked out the same way: each figure is one participant's series over its active window, from the turn it joined to the turn it left or the war or age ended, reduced by a mode chosen per figure.

- **Losses** (Strength Lost, Population, Crop and Production Lost): the sum of every turn-over-turn decline, with rises ignored so ordinary growth never hides a loss. Population Lost reads raw population, not the scaled chart value, so neither the people scaling nor the age-change smoothing on the Population chart can hide a real drop.
- **Net** (Settlements, Land): `last − first` over the window, signed (`+gained` / `−lost`). These count cities and territory that changed hands by capture, so a war from before the tracking began reads "-" rather than a misleading 0.
- **Accrued / Spent** (Settlements Razed, Production Directed to War, Refugees): the increase of a running counter over the window.
- **Level** (current Military Power): the last sampled value, a standing figure rather than a flow.
- **Casualties** (Military Strength Lost) use the engine's running units-killed counter when it is present, and fall back to the decline of the standing army for older saves.

A crisis's cumulative cost sums its stages. Because losses are sums of per-turn dips, they need dense samples, so each finished age's cumulative cost is captured when that age ends and stays accurate after old samples are thinned.

Sampling and the size of the save

Recorded history is capped automatically. The caps scale with game speed and adapt to the machine (processor cores, memory) and to the number of players, with a floor so history is never starved. Old samples are thinned first, and the

history is thinned further once its saved size passes 3 MB, so a long game with many civilizations cannot grow the save without limit. Charts also draw at most a set number of points per line, so a marathon game plots quickly.

Thinning is why a loss figure on a heavily thinned stretch can read "-" rather than a guess, while standing figures (current Military Power, counts) survive on a single sample.

A value written by a newer Demographics, or one this build cannot parse, is kept beside its key and restored the next time a build that can read it loads the save, so a save is safe to open in either build.

The Hall of Fame record

The Hall of Fame keeps one compact record per campaign in the shared `modSettings` entry, beside the settings, so it can be read from the main menu. Each save also carries its own campaign's record, which is added back to the Hall of Fame whenever that save is played, and recording the same game again updates its one entry rather than adding another.

Names are stored once for the whole archive rather than inside each game, so a full Hall of Fame keeps every game's timeline. Territory maps are the largest part, so they are kept for the most recently played games. Past that, older games give up their maps, then the other civilizations' timeline tracks, then their timelines, before any game is dropped; the game being played is always kept whole.

Known limits

- At 1280x720, the smallest supported resolution, the layout has reached its lower bound while the text has not, so some pages are tight: everything is reachable and readable, but column headings and long leader names can be cut off. Higher resolutions are unaffected.
- History starts when the mod is first active in a game. A game it is added to part-way through has no record of the turns before.
- Losses are inferred from sampled figures, not read from the game, so a loss on a heavily thinned stretch of a long game can be under-counted or read "-".

For mod authors: the companion API

Other mods can add to the screen through an optional, load-order-independent API on `globalThis.DemographicsMetricsAPI`. It does nothing unless another mod calls it, and Demographics takes no dependency on the caller.

- **`registerMetric(spec)`**: add a per-civilization line-chart metric that is sampled, stored and charted like the mod's own. A spec may include a `tooltipAttribution(ctx)` callback whose returned string is shown as a source line in that metric's tooltip.
- **`registerMetricToPage(pageId, metricId, afterMetricId?)`**: put the metric's tab on an existing page, for example next to Population.
- **`registerMetricGroup(spec)`**: show several related metrics behind one tab with up to two rows of pills (a metric row and a view or units row), charting one member at a time. Each member stays registered and sampled on its own.
- **`registerPanel(spec)`**: add a whole page, optionally with its own sub-tabs, whose body the other mod draws. The screen adds it as its own top-level tab and calls `spec.render(container, ctx, subId)` to fill it; the chart title, time range and CSV toolbar are left off these pages.
- **`registerHubPages(hubId, pages, opts?)`**: add pages to one of the hubs named in `HUB_IDS` (`statistics`, `migration`, `geopolitics`), after a given page or at the end.

Registrations made before the screen loads are queued on `DemographicsMetricsAPI.pending` and run when it starts. *Emigration* uses all of these: its **Emigration** tab's **Population & Migration** page charts Population, Population Share, Net Migration, Emigration, Immigration and Refugees, in scaled people or the game's own population numbers, with tooltips that break each flow down by cause and markers for wars and disasters, beside its own Network, Causes, My Cities, Diversity, Policies, Notifications and Guide pages.

Layout

demographics.modinfo	scripts, screens and text, in a main-menu group and a game group
ui/demographics-bootstrap.js	entry script: loads the dock button and the sampler
ui/demographics-options.js	the settings in Options, Mods → Demographics
ui/core/	settings, text, palette, font scale, tiers, storage repair
ui/core/settings-keeper.js	the Tower Settings Keeper, as in the standalone mod
ui/metrics/	the metric registry, formatting and the companion API
ui/sampler/	the per-turn sample of every civilization, wars, age changes
ui/storage/	recorded history in the save: format, loading, thinning, budget
ui/screen-demographics/	the screen: charts, views, rankings, camera, stylesheets
ui/history/	the History tab, the Hall of Fame and its main-menu screen
text/<lang>/ModText.xml	the text in each of 11 languages (see text/README.md)
images/demographics-icon.svg	the dock button icon
tests/	the tests `npm run verify` runs (tests/run-all.mjs)
types/	type declarations for the engine and the mod
devtools/	development-only views and probes, never shipped
docs/civ7-storage-bug.md	the game's storage bug and what Repair mod storage does
docs/screenshots/	the screenshots above, and the rest of the views
docs/steam-workshop-*.md	the Workshop page text, long and short (.txt copies to paste)
docs/workshop-preview.png	the mod icon and Workshop preview (from the .svg)
scripts/build_readme_pdf.sh	builds README.pdf from this file
scripts/deploy.mjs	copies the files that ship into the game's Mods folder
scripts/steam-changelog.mjs	renders CHANGELOG.md into CHANGELOG.steam.txt
release.sh	checks the build, then writes the release zip into dist/
CHANGELOG.md	what changed in each version
CONTRIBUTING.md	setup, the checks to run and the code conventions
README.pdf	this file, typeset

Credits

Built by Tower. The History tab and the Hall of Fame come from Tower's History & Rankings mod.

Demographics follows robk's InfoAddict for Civilization V and Gedemon's CivGraphs for Civilization VI, and builds on Slothoth's Global Relations panel for Civilization VII. Thanks to Slothoth also for the keys to data persistence across ages, to AndySafik for the Polish translation and playtest feedback, to renouf for bug reports, and to the Civilization modding community for documentation, samples and testing.

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